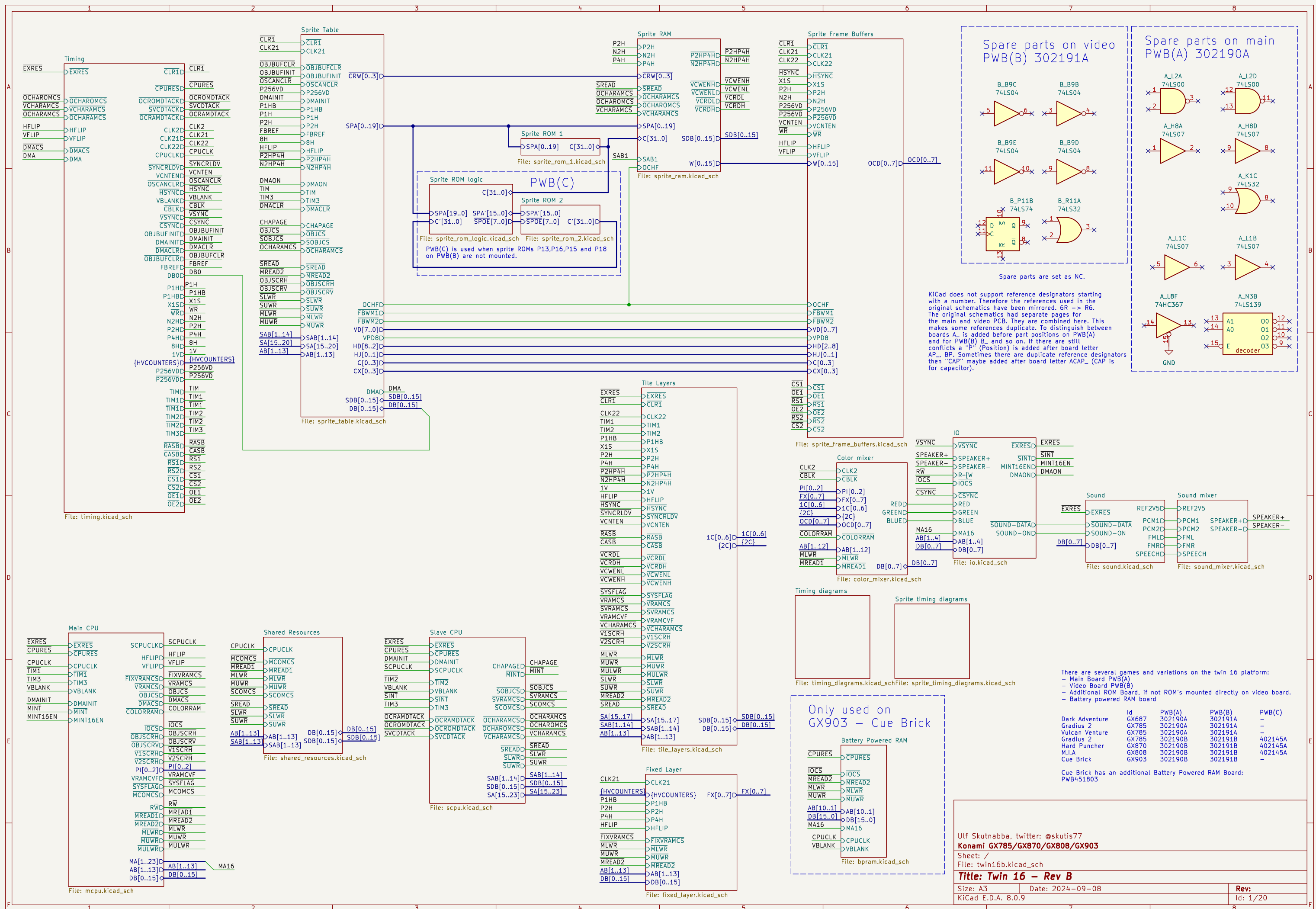
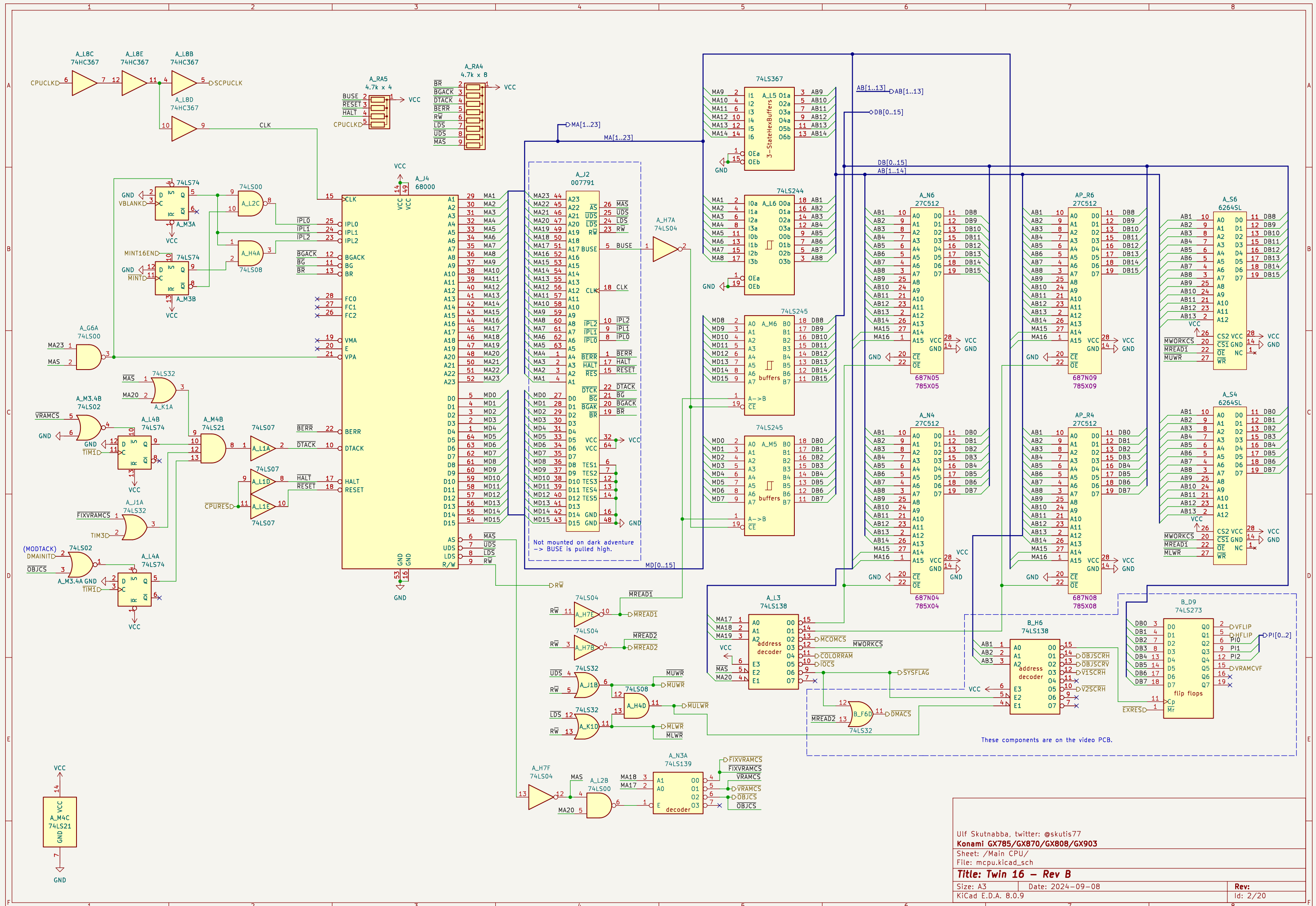
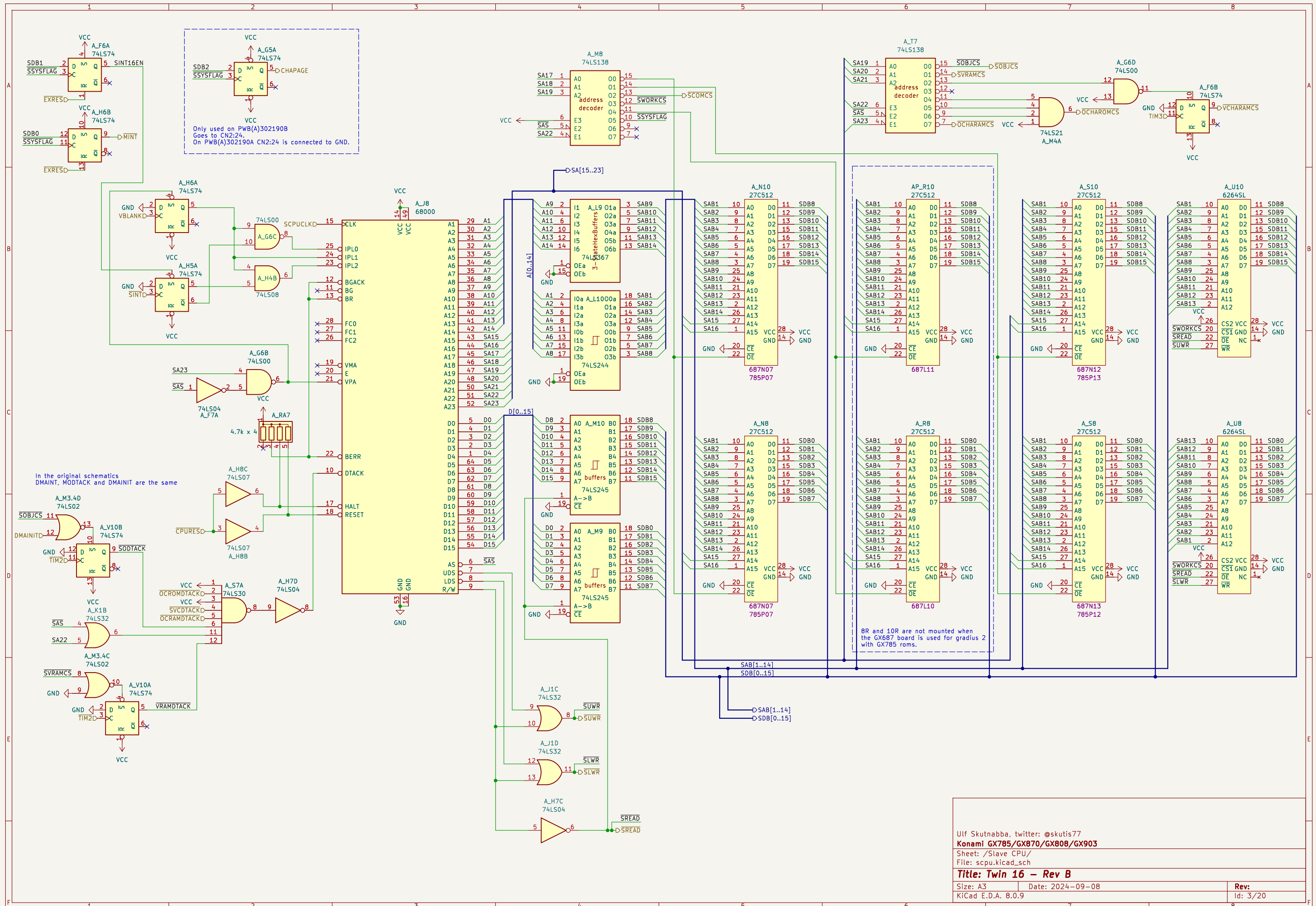




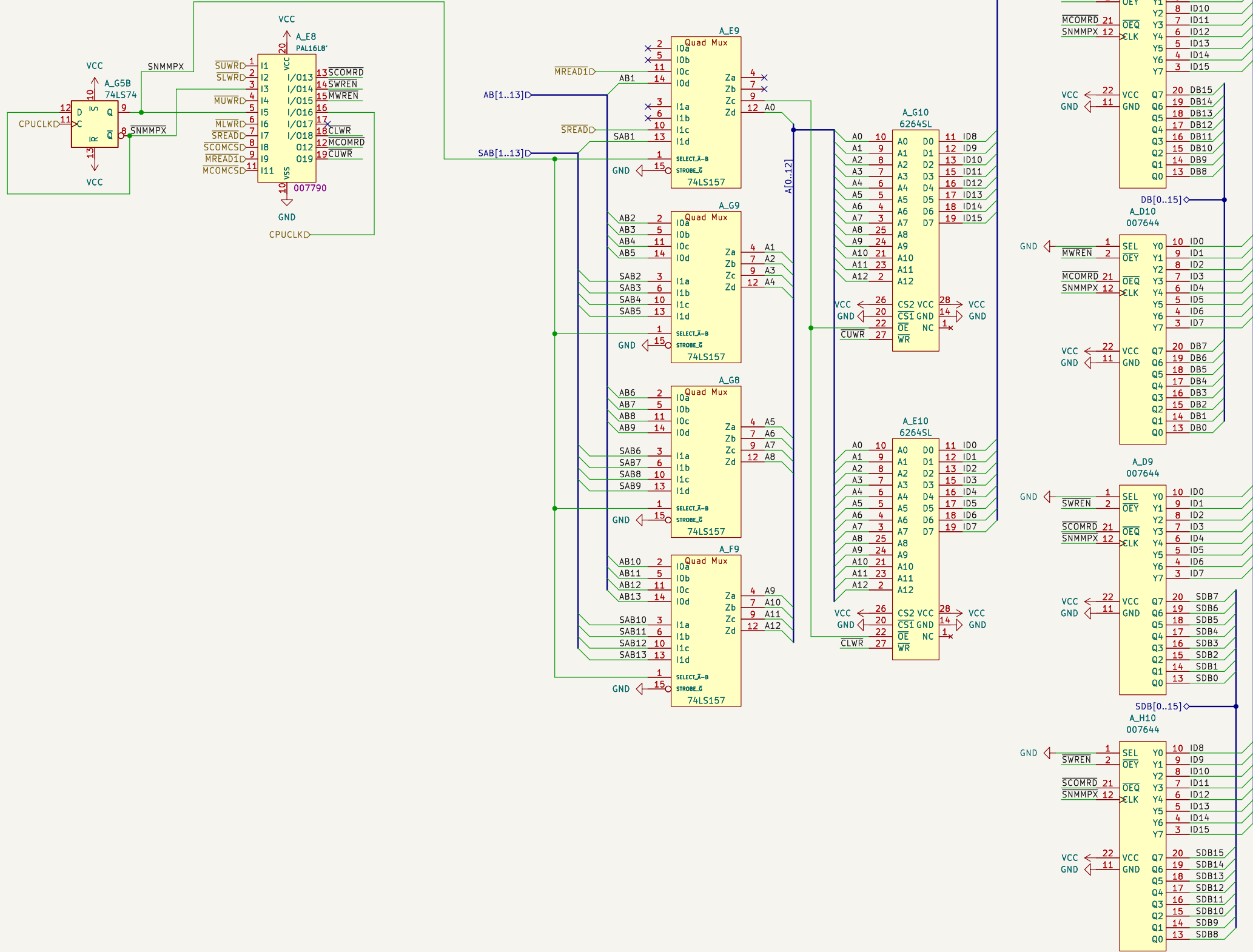




Konami 007790

```
MCOMRD = ~(-MREAD1 & ~MCOMCS) = MREAD1 | MCOMCS
SCOMRD = ~(-SREAD & ~SCOMCS)
SWREN = ~(-SNMMPX & SREAD & ~SCOMCS)
MWREN = ~(-SNMMPX & MREAD1 & ~MCOMCS)
CLWR = ~(-SLWR & ~SNMMPX & SNMMPX & SREAD & ~SCOMCS & ~CPUCLK | ~SNMMPX & ~MLWR & MREAD1 & ~MCOMCS & ~CPUCLK)
CUWR = ~(-SUWR & ~SNMMPX & SNMMPX & SREAD & ~SCOMCS & ~CPUCLK | ~MUWR & ~SNMMPX & MREAD1 & ~MCOMCS & ~CPUCLK)
```

All AND logic can be converted to OR form. But since AND logic and inverters are used in the PAL16L8 device it is kept in that form. The equations follow verilog syntax. The active low signals are not inverted, the bar is there only to show activeness.



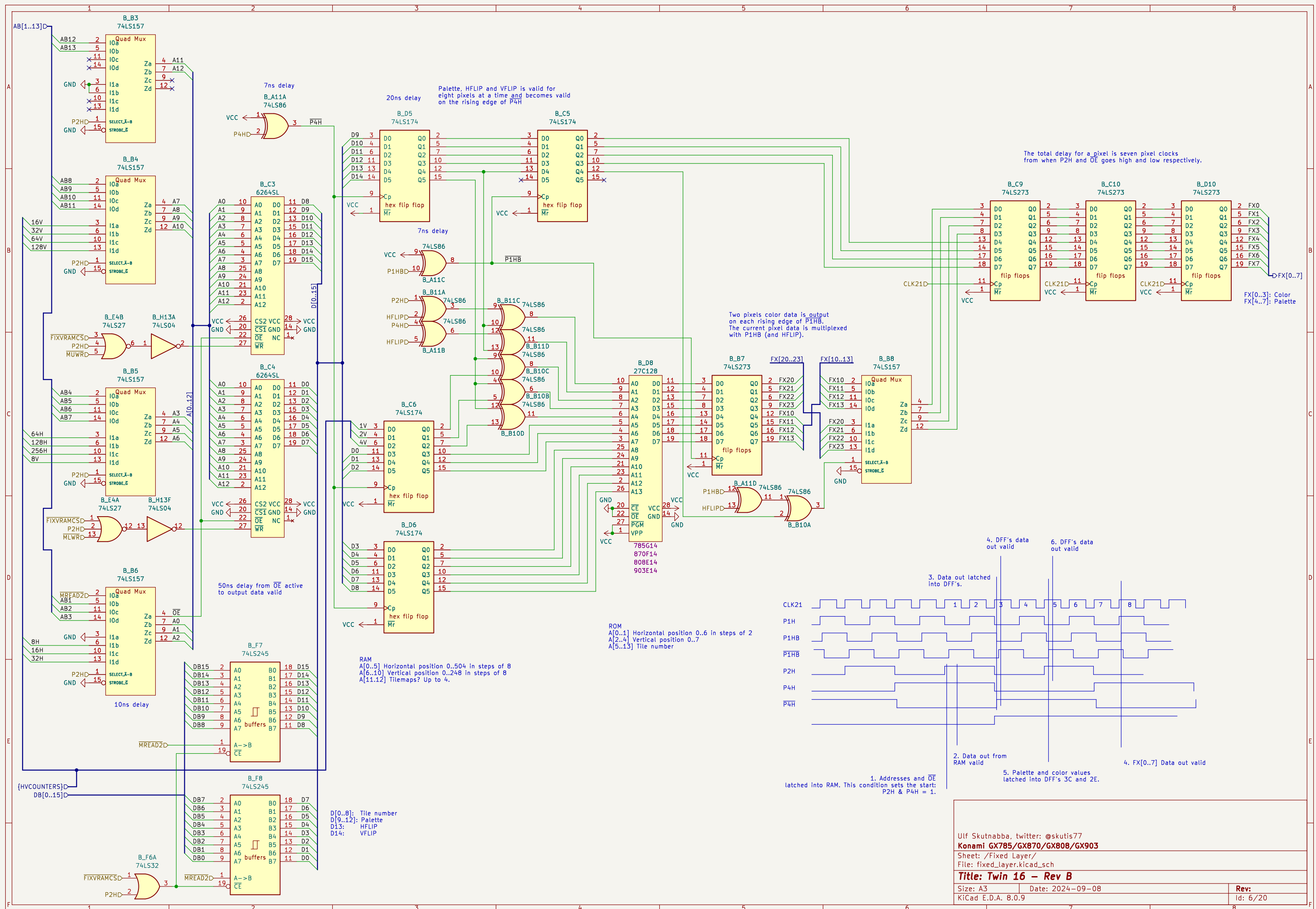
This is from jammarcade.net

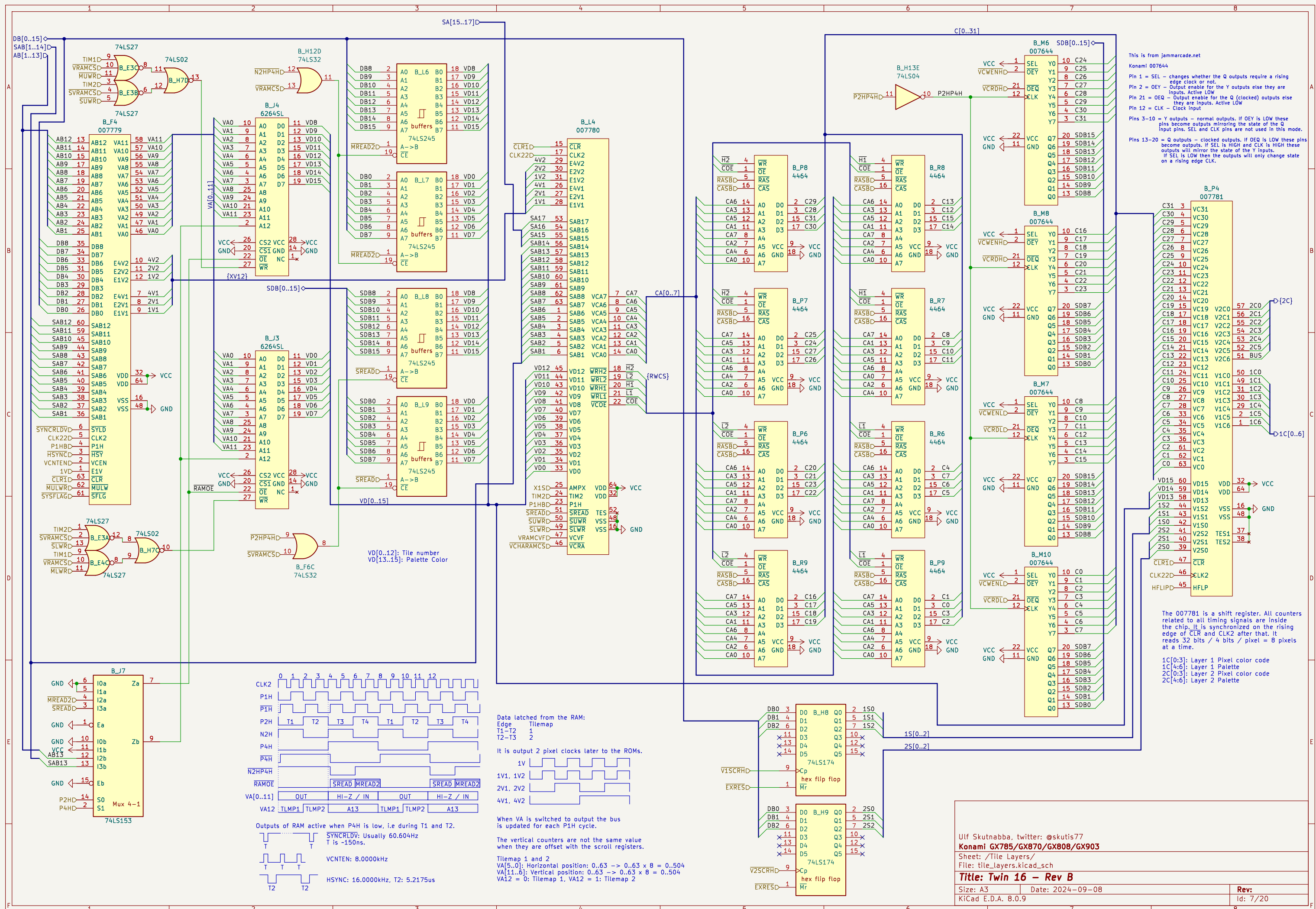
Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

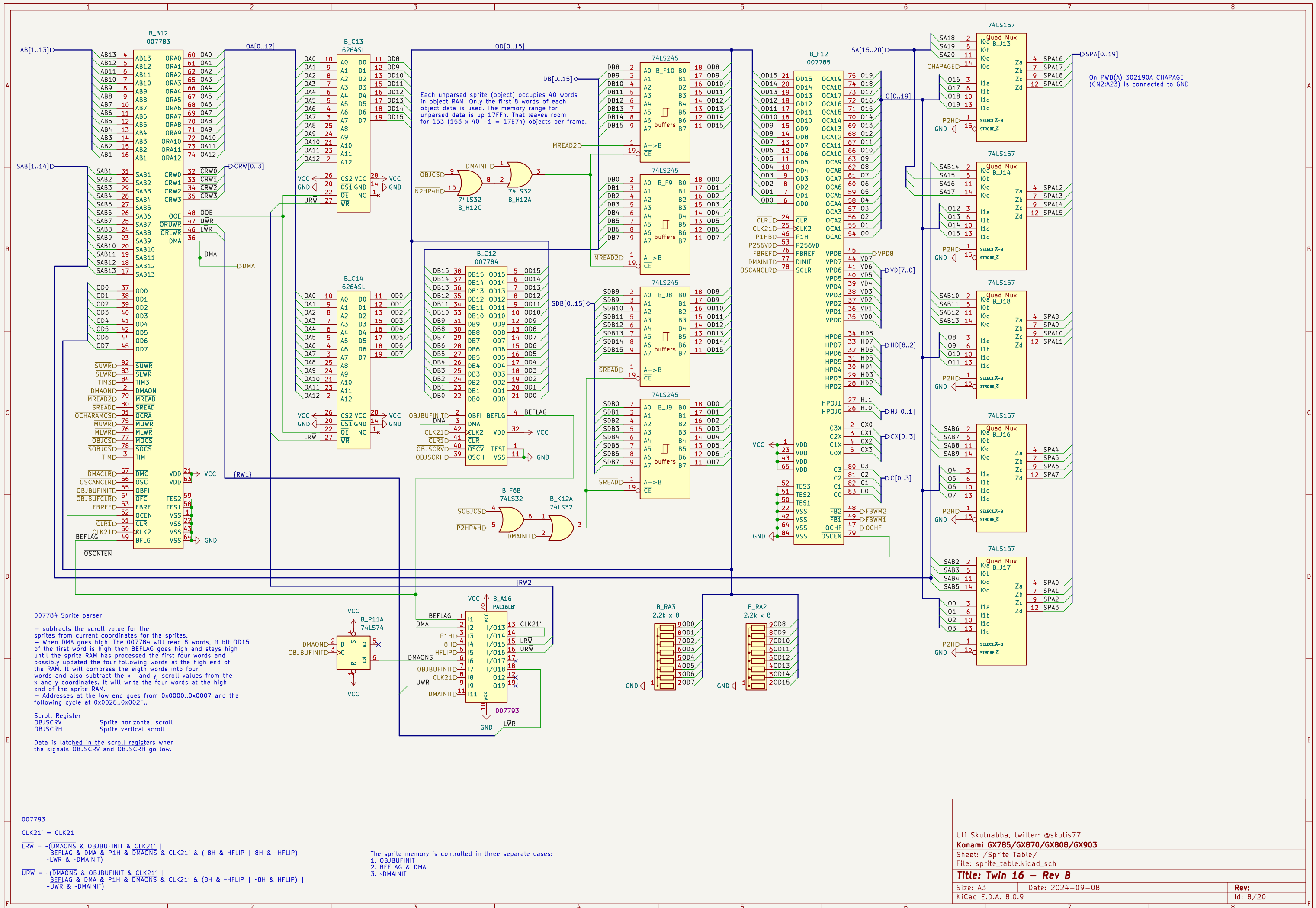
Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.









Graphics ROM and RAM

ROM naming	ID	P13	P15	P16	P18
cuebrickj	GX903	NP	NP	NP	NP
miaj	GX808	808D15	NP	808D17	NP
hpuncher	GX870	870C15	870C16	870C17	870C18
gradius2 / vulcan	GX785	785F15	785F16	785F17	785F18

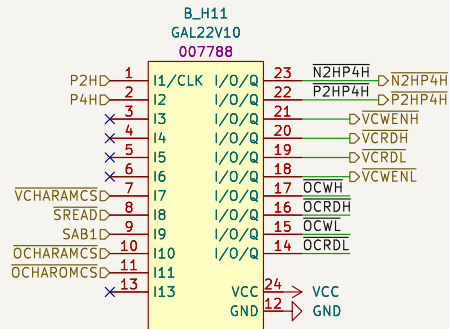
NP = Not Populated
On some boards P13,P15,P16 and P18 are NP and replaced with PWB(C) 402145A. See separate schematics.

This is from jammarcade.net

Pin 1 = SEL – changes whether the Q outputs require a rising edge clock or not.
Pin 2 = OEY – Output enable for the Y outputs else they are inputs. Active LOW
Pin 21 = OEQ – Output enable for the Q (clocked) outputs else they are inputs. Active LOW
Pin 12 = CLK – Clock input

Pins 3–10 = Y outputs – normal outputs. If OEY is LOW these pins become outputs mirroring the state of the Q input pins. SEL and CLK pins are not used in this mode.

Pins 13–20 = Q outputs – clocked outputs. If OEQ is LOW these pins become outputs. If SEL is HIGH and CLK is HIGH these outputs will mirror the state of the Y inputs. If SEL is LOW then the outputs will only change state on a rising edge CLK.



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The inverted -P2H is written as N2H.

OCRD1 = ~(-SREAD & SAB1 & ~OCHARAMCS & OCHAROMCS) | ~SREAD & SAB1 & ~OCHAROMCS
OCWL = ~(-P2H & SREAD & SAB1 & ~OCHARAMCS)
OCRDH = ~(-SREAD & ~SAB1 & ~OCHARAMCS & OCHAROMCS) | ~SREAD & ~SAB1 & ~OCHAROMCS
OCWH = ~(-P2H & SREAD & SAB1 & ~OCHARAMCS)
VCWENL = ~(-P2H & P4H & ~VCHARAMCS & SREAD & SAB1)
VCRDL = ~(-VCHARAMCS & ~SREAD & SAB1)
VCRDH = ~(-VCHARAMCS & ~SREAD & ~SAB1)
VCWENH = ~(-P2H & P4H & ~VCHARAMCS & SREAD & ~SAB1)
P2HP4H = ~(-P2H & P4H)
N2HP4H = ~(-P2H & P4H)

C[31..0]

Sprite Character Banks

Page	SPA19	SPA18	SPA17
1	X	L	L
2	X	L	L
3	H	H	L
4	H	H	L
5	RAM	H	H

CRW[0..3]D

B_H15C 74LS00

SPA17 9

SPA18 10

B_M15 SRAM32K8B

SPA0	10	A0	D0	11	C24
SPA1	9	A1	D1	12	C25
SPA2	8	A2	D2	13	C26
SPA3	7	A3	D3	15	C27
SPA4	6	A4	D4	16	C28
SPA5	5	A5	D5	17	C29
SPA6	4	A6	D6	18	C30
SPA7	3	A7	D7	19	C31
SPA8	25	A8			
SPA9	24	A9			
SPA10	21	A10			
SPA11	23	A11			
SPA12	2	A12			
SPA13	26	A13			
SPA14	1	A14			

GND	20	CS	VCC	28	VCC
RAMOE	22	OE	VSS	14	GND
CRW3	27	WR			

B_M14 SRAM32K8B

SPA0	10	A0	D0	11	C16
SPA1	9	A1	D1	12	C17
SPA2	8	A2	D2	13	C18
SPA3	7	A3	D3	15	C19
SPA4	6	A4	D4	16	C20
SPA5	5	A5	D5	17	C21
SPA6	4	A6	D6	18	C22
SPA7	3	A7	D7	19	C23
SPA8	25	A8			
SPA9	24	A9			
SPA10	21	A10			
SPA11	23	A11			
SPA12	2	A12			
SPA13	26	A13			
SPA14	1	A14			

GND	20	CS	VCC	28	VCC
RAMOE	22	OE	VSS	14	GND
CRW2	27	WR			

B_M19 SRAM32K8B

SPA0	10	A0	D0	11	C8
SPA1	9	A1	D1	12	C9
SPA2	8	A2	D2	13	C10
SPA3	7	A3	D3	15	C11
SPA4	6	A4	D4	16	C12
SPA5	5	A5	D5	17	C13
SPA6	4	A6	D6	18	C14
SPA7	3	A7	D7	19	C15
SPA8	25	A8			
SPA9	24	A9			
SPA10	21	A10			
SPA11	23	A11			
SPA12	2	A12			
SPA13	26	A13			
SPA14	1	A14			

GND	20	CS	VCC	28	VCC
RAMOE	22	OE	VSS	14	GND
CRW1	27	WR			

B_M17 SRAM32K8B

SPA0	10	A0	D0	11	C0
SPA1	9	A1	D1	12	C1
SPA2	8	A2	D2	13	C2
SPA3	7	A3	D3	15	C3
SPA4	6	A4	D4	16	C4
SPA5	5	A5	D5	17	C5
SPA6	4	A6	D6	18	C6
SPA7	3	A7	D7	19	C7
SPA8	25	A8			
SPA9	24	A9			
SPA10	21	A10			
SPA11	23	A11			
SPA12	2	A12			
SPA13	26	A13			
SPA14	1	A14			

GND	20	CS	VCC	28	VCC
RAMOE	22	OE	VSS	14	GND
CRW0	27	WR			

Pixel color data output rate is:
16-bits / 2 pixels = 8 bits / pixel.

W[15]: Vertical flip

B_P19 74LS374

C24	3	D0	00	2	W8
C25	4	D1	01	5	W9
C26	7	D2	02	6	W10
C27	8	D3	03	9	W11
C28	13	D4	04	12	W12
C29	14	D5	05	15	W13
C30	17	D6	06	16	W14
C31	18	D7	07	19	W15

3-state flip flops

N2H 11

OE 1

CP 07

W[0..15]

B_R19 74LS374

C16	3	D0	00	2	W0
C17	4	D1	01	5	W1
C18	7	D2	02	6	W2
C19	8	D3	03	9	W3
C20	13	D4	04	12	W4
C21	14	D5	05	15	W5
C22	17	D6	06	16	W6
C23	18	D7	07	19	W7

3-state flip flops

N2H 11

OE 1

CP 07

W[0..15]

B_P20 74LS374

C8	3	D0	00	2	W8
C9	4	D1	01	5	W9
C10	7	D2	02	6	W10
C11	8	D3	03	9	W11
C12	13	D4	04	12	W12
C13	14	D5	05	15	W13
C14	17	D6	06	16	W14
C15	18	D7	07	19	W15

3-state flip flops

N2H 11

OE 1

CP 07

W[0..15]

B_R20 74LS374

C0	3	D0	00	2	W0
C1	4	D1	01	5	W1
C2	7	D2	02	6	W2
C3	8	D3	03	9	W3
C4	13	D4	04	12	W4
C5	14	D5	05	15	W5
C6	17	D6	06	16	W6
C7	18	D7	07	19	W7

3-state flip flops

N2H 11

OE 1

CP 07

W[0..15]

B_H13C 74LS04

5

6

Ulf Skutnabba, twitter: @skutis77

Konami GX785/GX870/GX808/GX903

Sheet: /Sprite RAM/

File: sprite_ram.kicad_sch

Title: Twin 16 – Rev B

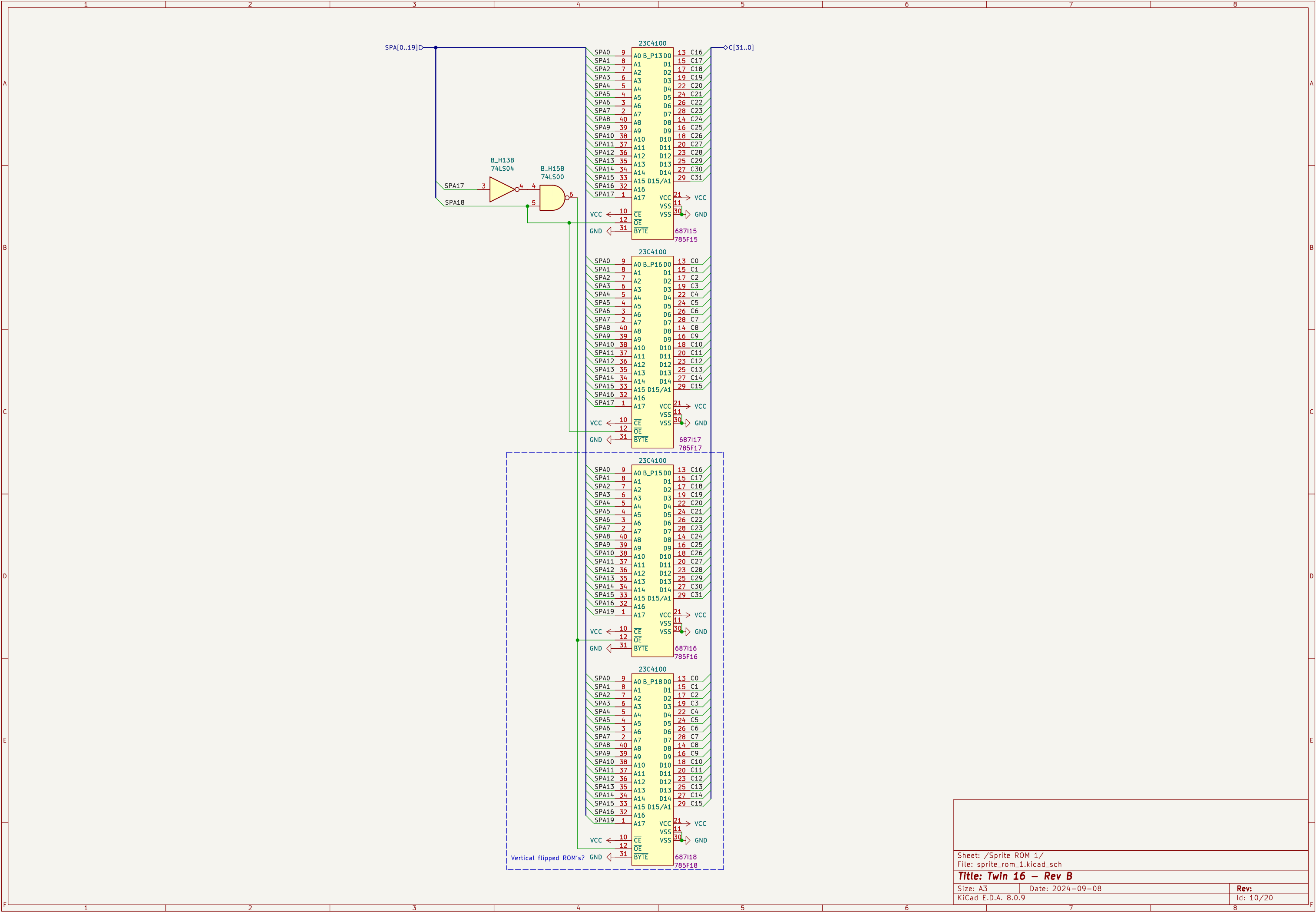
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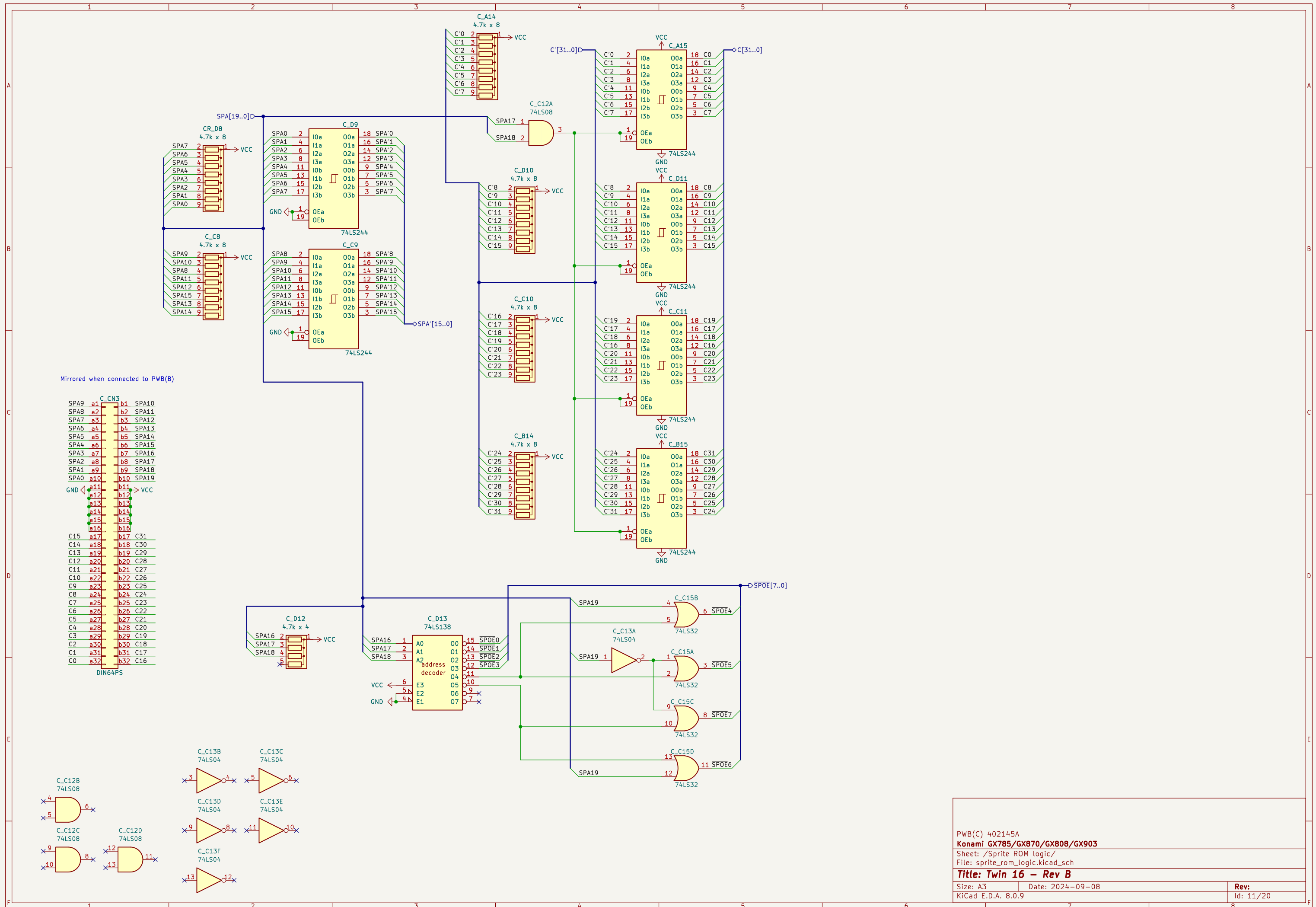
Date: 2024–09–08

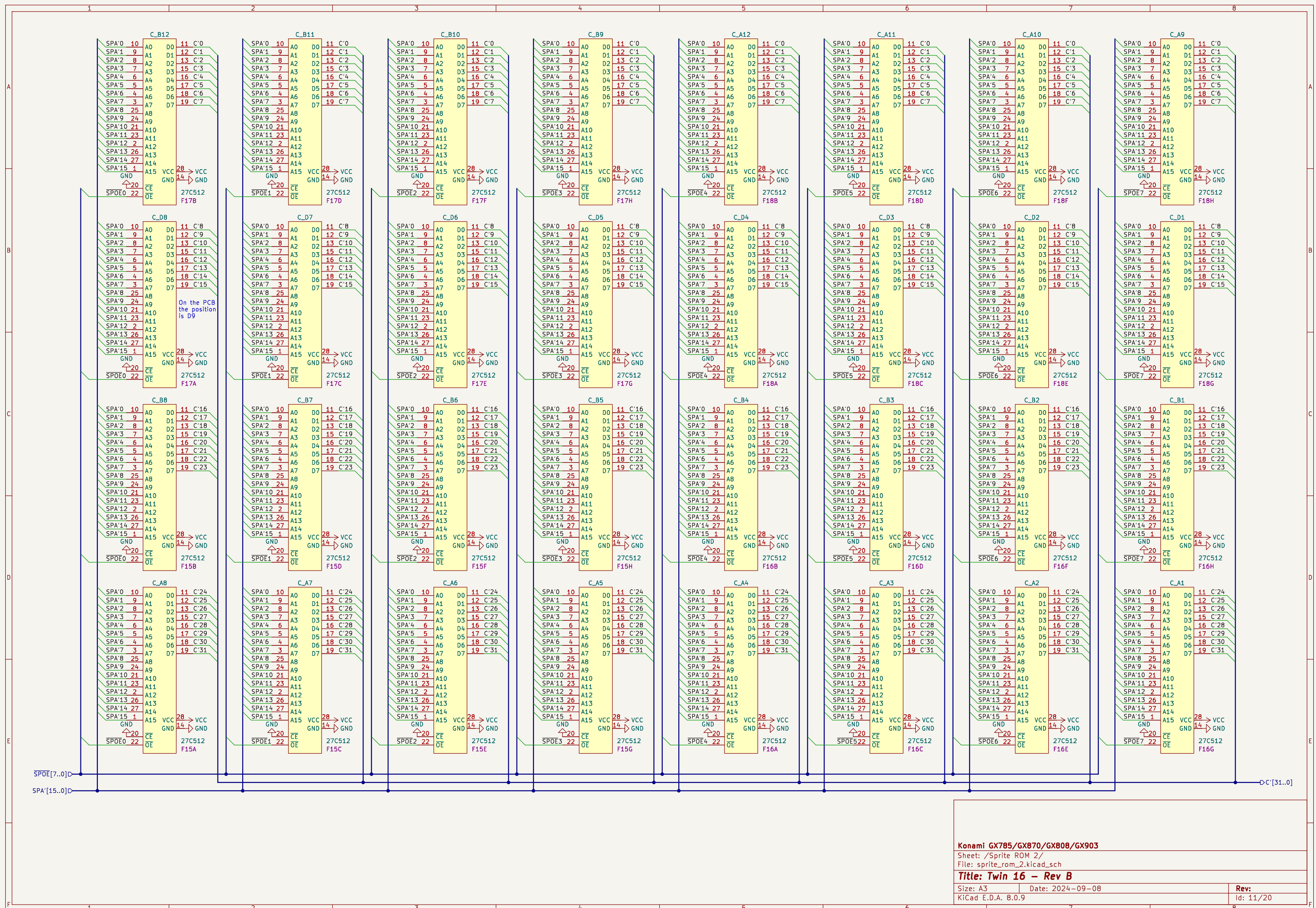
Rev:

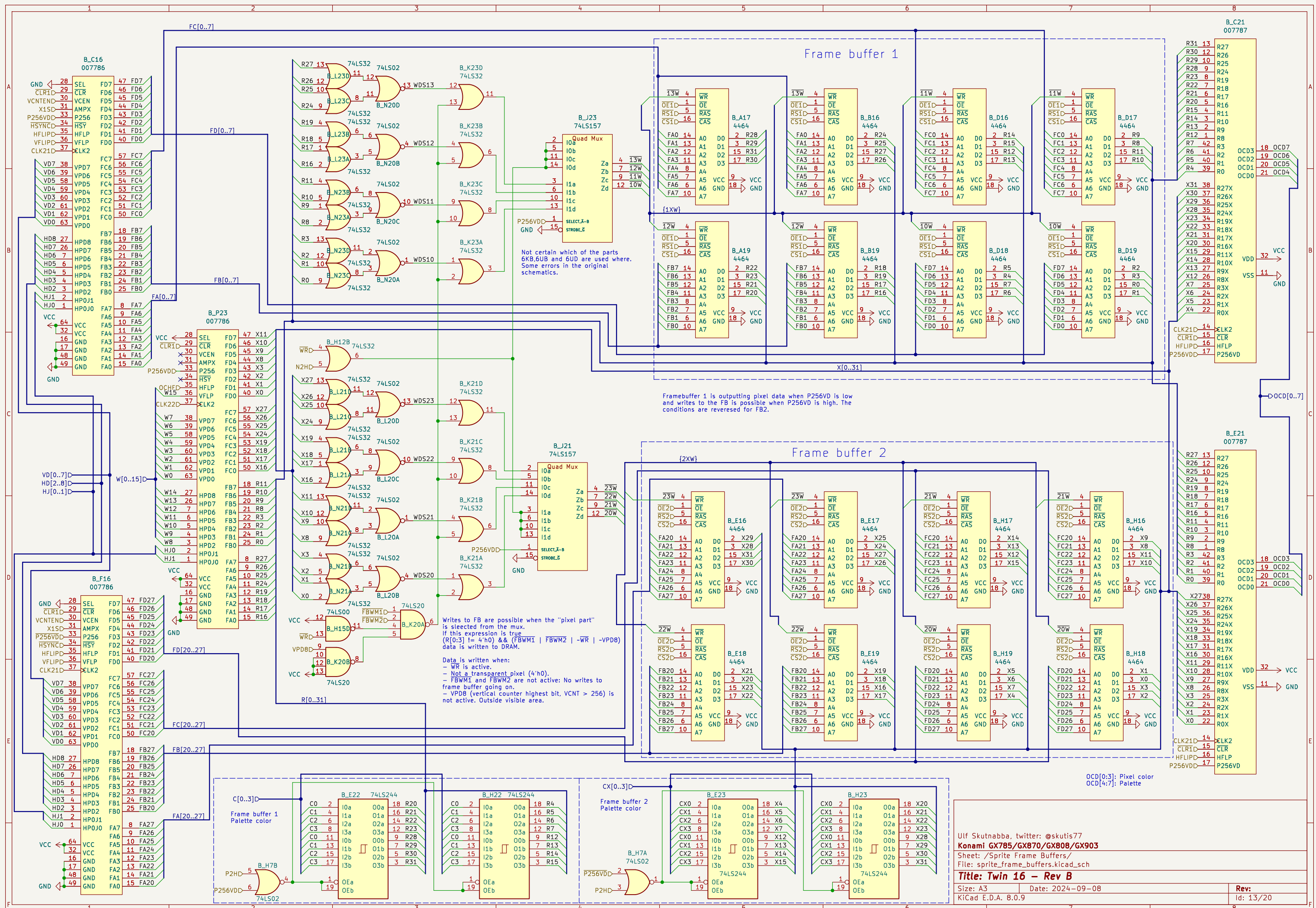
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Id: 9/20



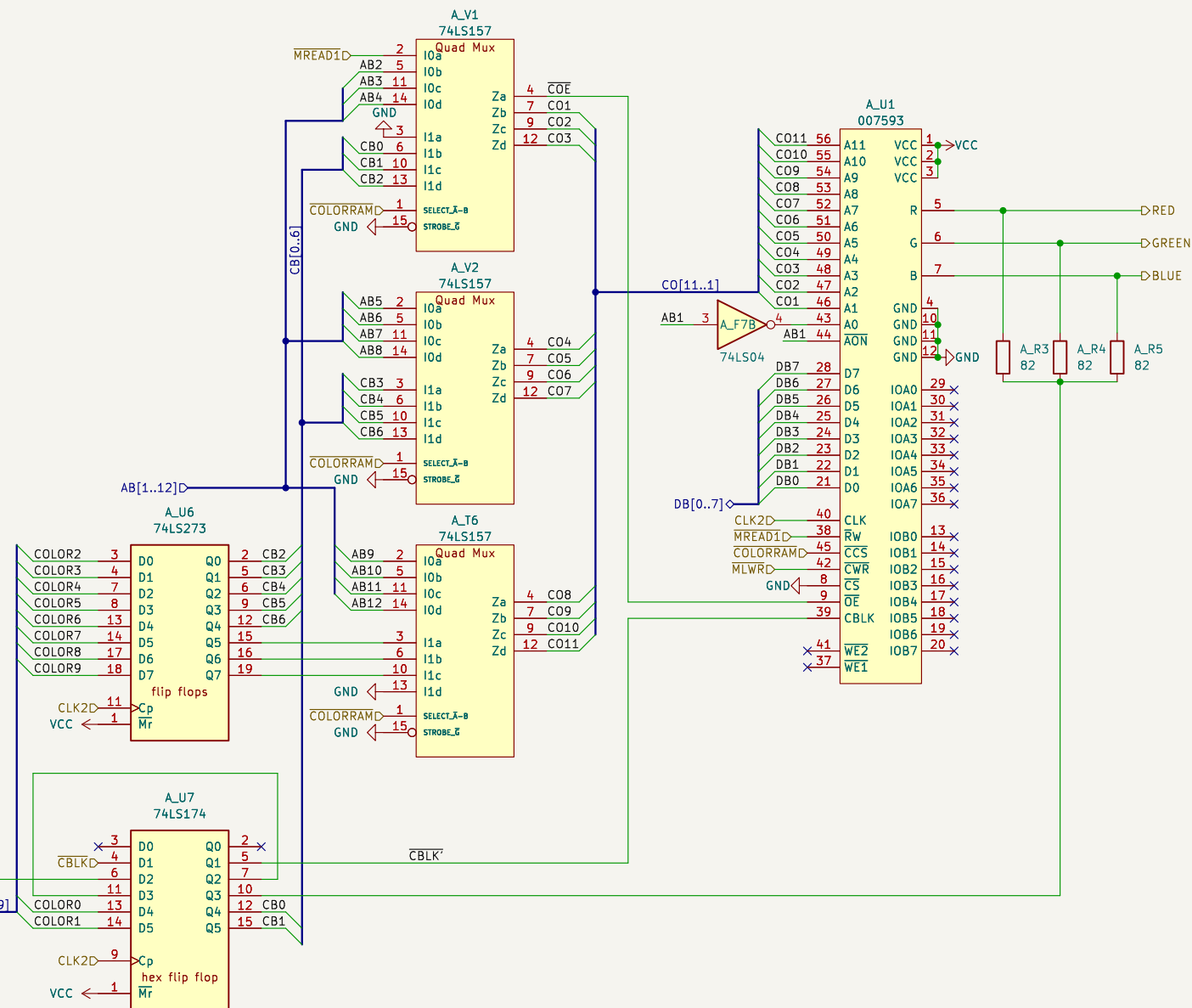
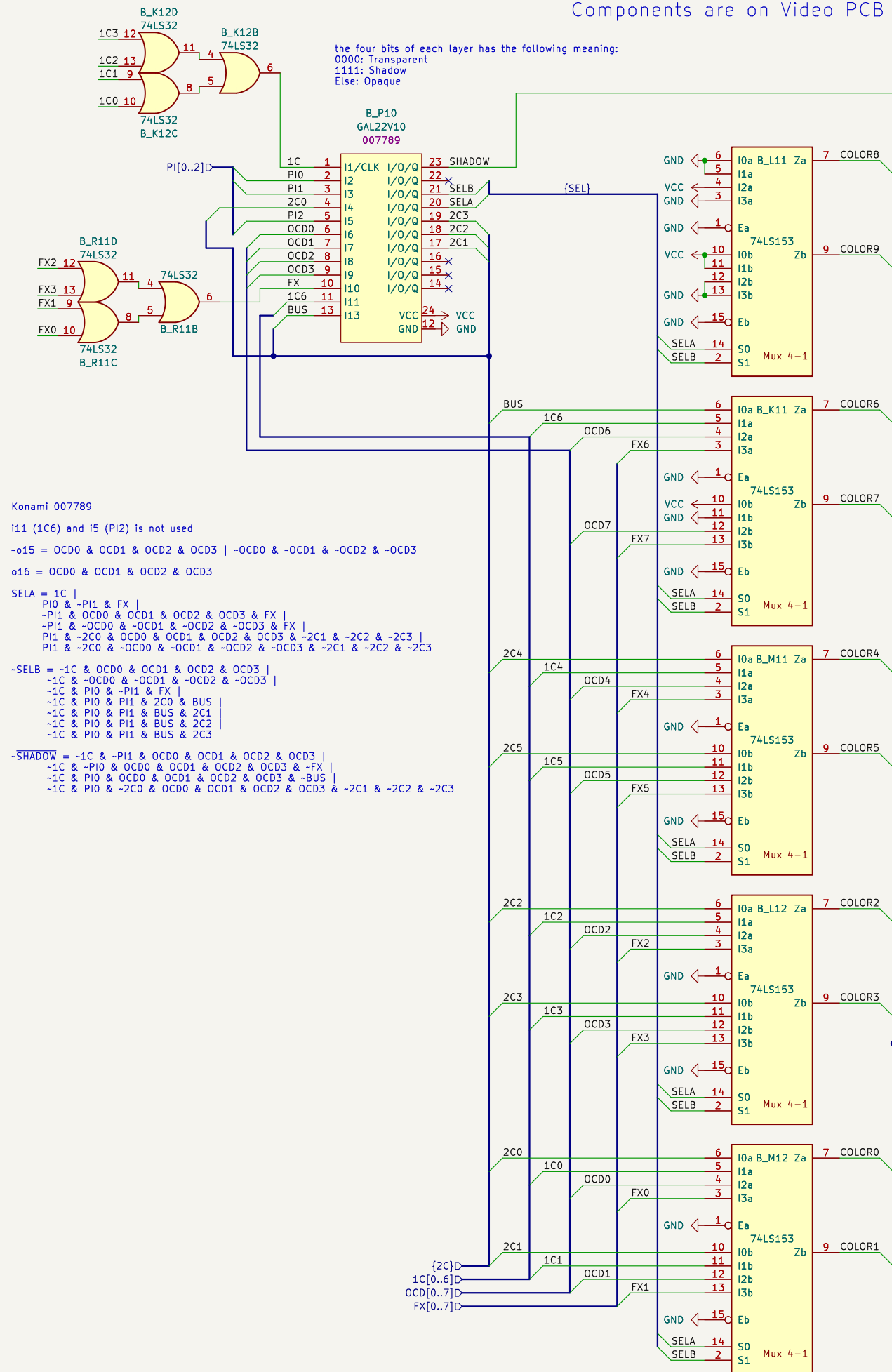


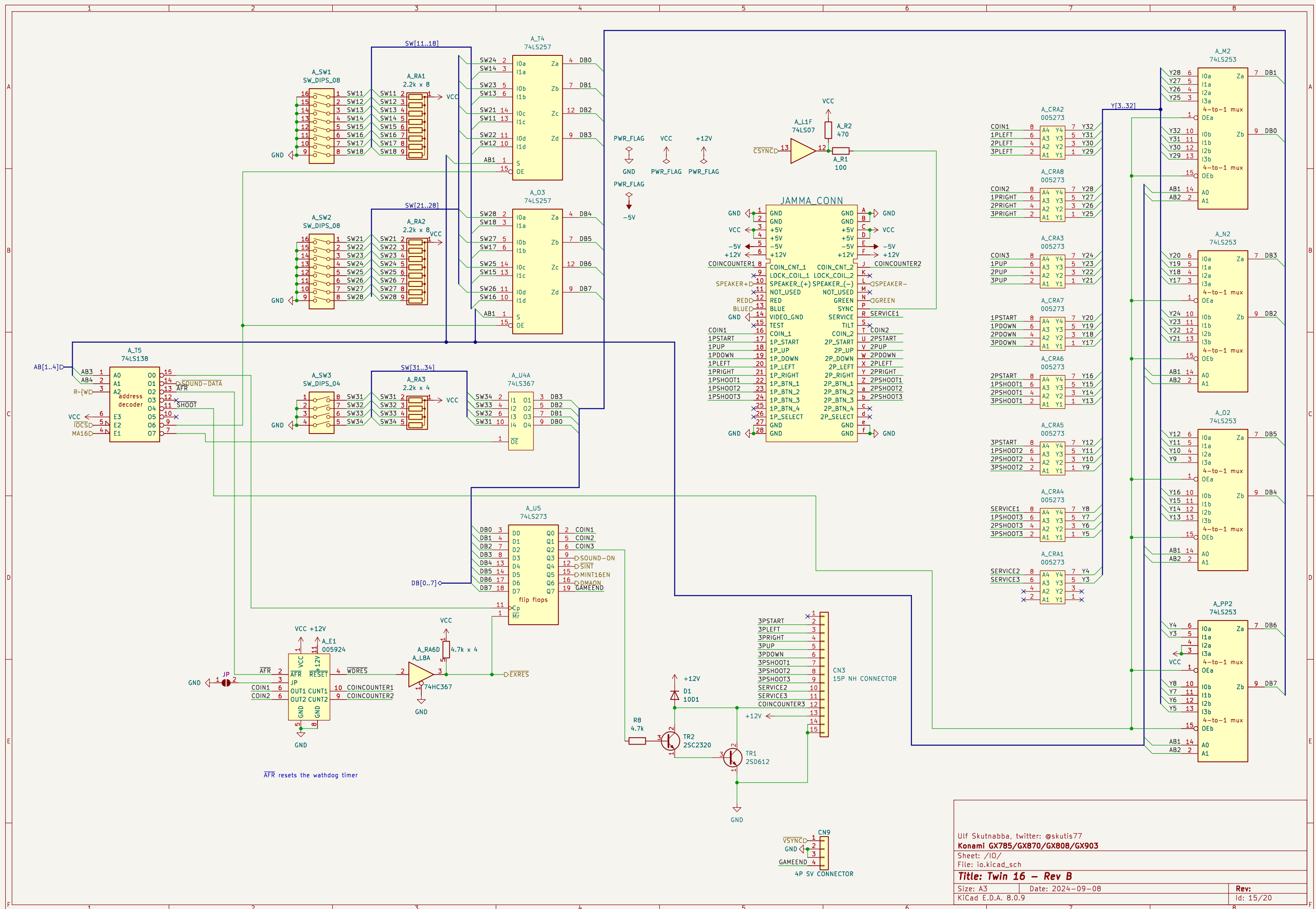


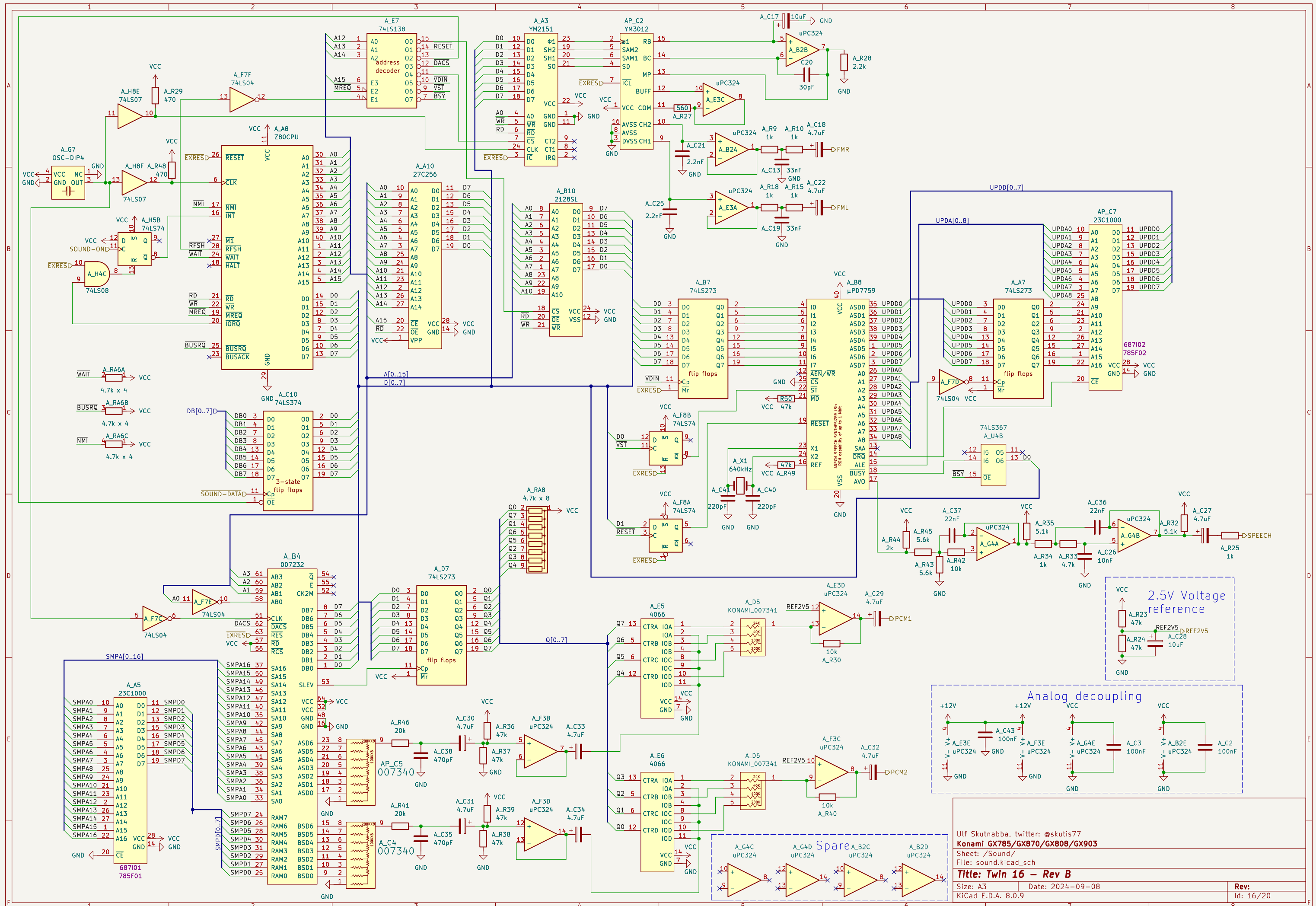


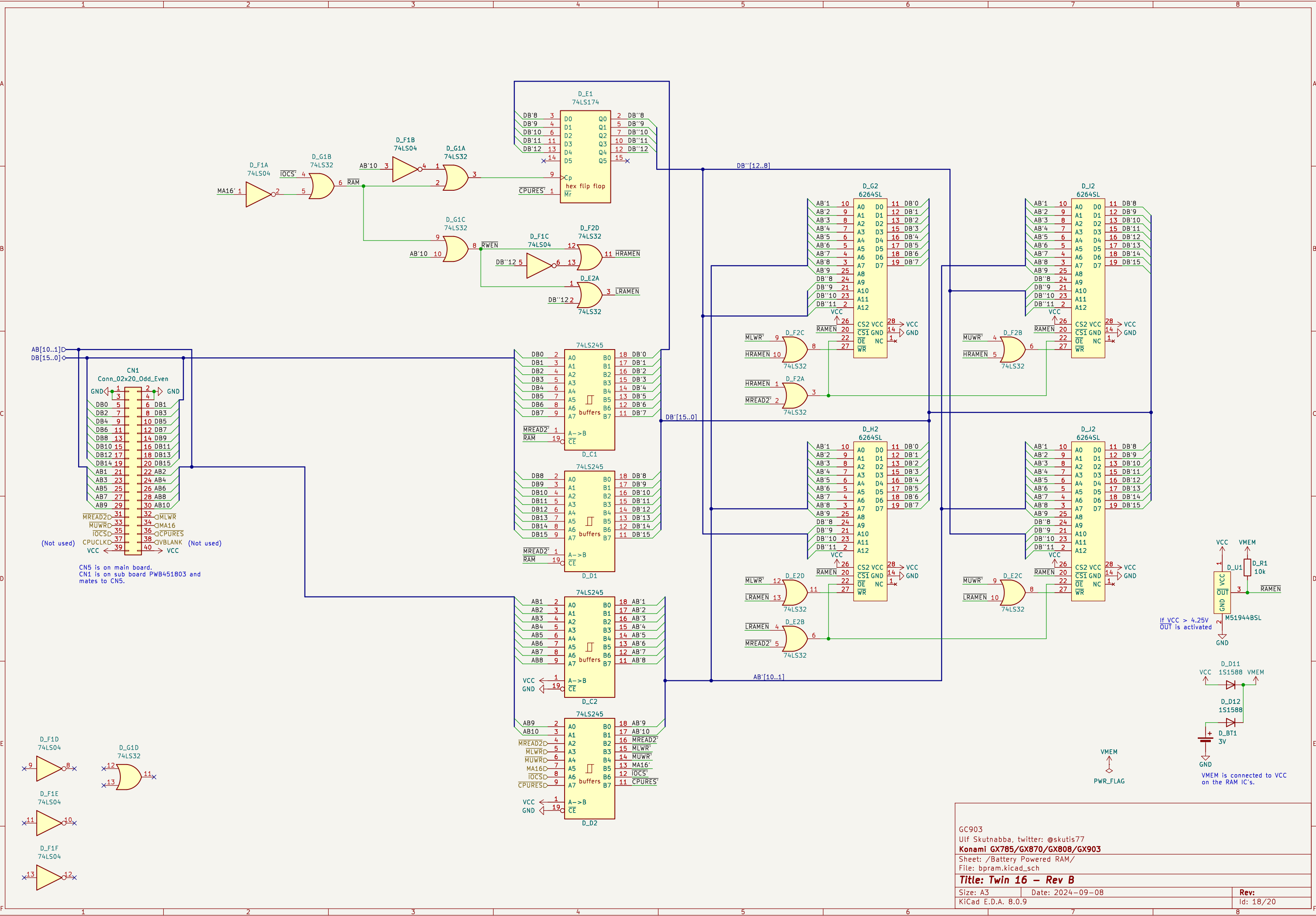
Components are on Video PCB

Components are on Main PCB

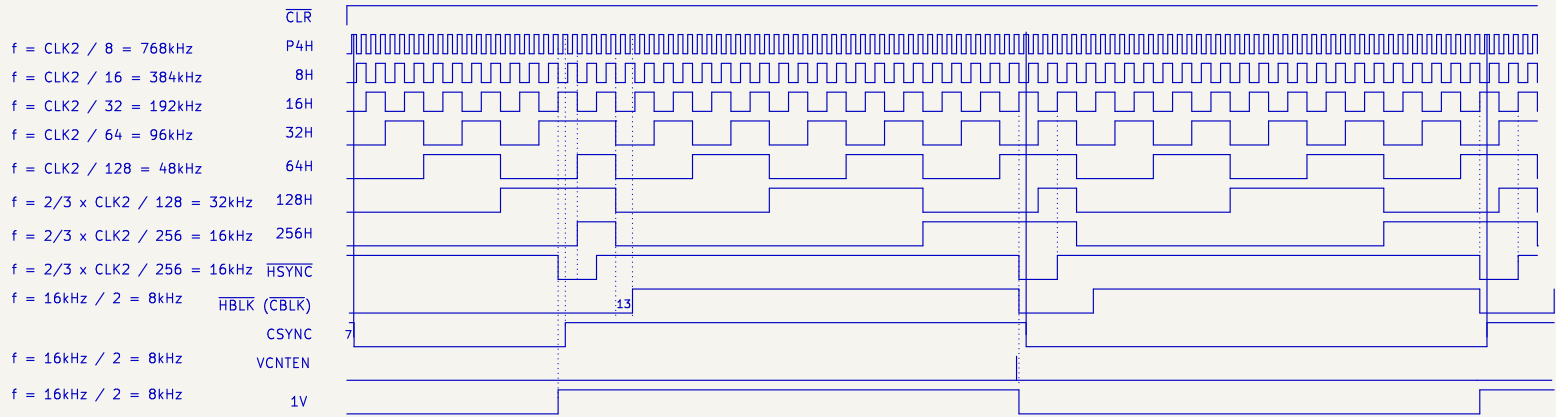








Horizontal signals

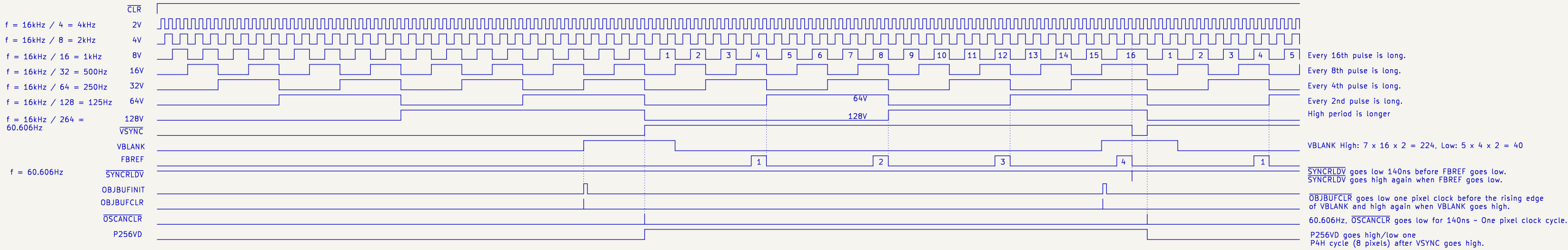
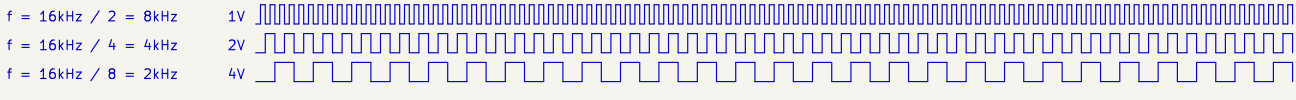


– The first VCNTEN is skipped after reset. It goes low 140ns before HSYNC goes low, and high again when HSYNC goes low. VCNTEN is active right before every second falling edge of HSYNC.

– CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

The first HSYNC is at HCOUNT 175 after reset.

Vertical signals



Every 16th pulse is long.
Every 8th pulse is long.
Every 4th pulse is long.
Every 2nd pulse is long.
High period is longer

VBLANK High: $7 \times 16 \times 2 = 224$, Low: $5 \times 4 \times 2 = 40$

SYNCRLDV goes low 140ns before FBREF goes low. SYNCRLDV goes high again when FBREF goes low.

OBJBUFCLR goes low one pixel clock before the rising edge of VBLANK and high again when VBLANK goes high.

60.606Hz, OSCANCLR goes low for 140ns – One pixel clock cycle.

P256VD goes high/low one P4H cycle (8 pixels) after VSYNC goes high.

FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

Horizontal and vertical synch timing diagrams

